

# VIT Bhopal University

Bhopal, Madhya Pradesh

## PROJECT REPORT

# Rock, Paper, Scissors

Command-Line Game in Python

|                  |                      |
|------------------|----------------------|
| Name             | K Aditya Prakash     |
| Registration No. | 26BSA10013           |
| Programme        | BTech CSE 1021       |
| Faculty          | Pradeep Kumar Mishra |
| Language         | Python               |
| Date             | 29/09/2026           |

# Project Report

Rock, Paper, Scissors (CLI Game)

## 1. Introduction / Problem Statement

A command-line game where a user plays three rounds of Rock, Paper, Scissors against a computer opponent. The project practices core Python concepts: user input, randomness, conditionals, loops and functions.

## 2. Objectives

- Build a fully terminal-based interactive game.
- Implement standard game rules and score tracking.
- Determine and display an overall winner after three rounds.

## 3. Technologies Used

Python 3.6+ and the standard random module. There are no external dependencies.

## 4. Design and Implementation

- `game()` initializes player and computer scores at 0.
- A for loop runs 3 rounds. Each round reads the player's choice with `input()` and picks the computer's choice with `random.choice()`.
- An `if/elif/else` chain compares the two choices: a tie awards no points, rock beats scissors, paper beats rock, and scissors beats paper. Any other outcome goes to the computer.
- After the loop, the program compares the scores and prints the winner, or a tie.

## 5. How to Run

See README.md: check Python is installed, clone the repo, then run:

```
python rock_paper_scissors.py
```

## 6. Sample Output

```
Round 1
Select rock, paper, or scissors: rock
Computer chose: scissors
You win!
Round 2
Select rock, paper, or scissors: paper
Computer chose: paper
It's a tie!
Round 3
Select rock, paper, or scissors: scissors
Computer chose: rock
Computer wins!
Final Score:
Your score: 1
Computer score: 1
It's a tie game!
```

## 7. Testing

Tested manually by playing full games and checking that a win, loss and tie each score correctly, that the final result matches the scores, and that all 9 choice combinations resolve properly.

## 8. Limitations and Future Improvements

- Input is case-sensitive, and invalid input counts as a computer win. This could be fixed with `.strip().lower()` and re-prompting.
- The number of rounds is fixed at 3 and could be made configurable.
- Possible additions are a play-again option, score history and a best-of-N mode.

## 9. Conclusion

The project meets its goals: a working, dependency-free CLI game that follows the standard rules and reports a clear result.

## 10. Repository Link

<https://github.com/kadityaparakash009/RockPaperScissorsVit>